Grace — θ12 cleanup: compute $|$U_e2$|^2$ forward — the matrix-element form is N_c/(rank·n_C)=3/10; "42/137 BANKED" is over-claimed

2026-07-14 Tuesday. Keeper cleanup (mine): θ12 is over-claimed in the scorecard ("BANKED, sin$^2\theta$12=42/137, 0.14%") — it papers over a derived matrix-element form with a mechanism vs a selected number. Compute $|$U_e2$|^2$ forward to settle it cheaply, no experiment. DONE: the physical observable is $|$U_e2$|^2$ (not sin$^2\theta$12), and its cleanest BST form is N_c/(rank·n_C)=3/10 (0.06%) — 42/137 is a near-coincident fit to the derived-then-divided sin$^2\theta$12. θ12 → IDENTIFIED, not BANKED.

The physical observable is the MATRIX ELEMENT $|$U_e2$|^2$, not sin$^2\theta$12

Oscillation data measures the PMNS matrix elements. $|$U_e2$|^2$ = sin$^2\theta$12·cos$^2\theta$13. With sin$^2\theta$13 = 1/45 (banked):
- observed $|$U_e2$|^2$ = 0.307 × (44/45) = 0.3002.

Compute forward — the cleanest form is the matrix element

form	value	vs $|$U_e2$|^2$(0.3002)	vs sin$^2\theta$12(0.307)	for
N_c/(rank·n_C) = 3/10	0.3000	0.06% ▯	2.28%	$|$U_e2$|^2$
(C_2·g)/N_max = 42/137	0.3066	2.13%	0.14%	sin$^2\theta$12
rank^2/c_3 = 4/13	0.3077	2.50%	0.23%	sin$^2\theta$12
27/88 = (3/10)/(44/45)	0.3068	2.21%	0.06%	sin$^2\theta$12

The chain: $|$U_e2$|^2$ = N_c/(rank·n_C) = 3/10 (mechanism-shaped: color / (rank·dim); matches the physical matrix element at 0.06%) ▯ sin$^2\theta$12 = (3/10)/cos$^2\theta$13 = (3/10)·(45/44) = 135/440 = 27/88 = 0.3068. And 42/137 = 0.3066 is a numerical near-coincidence with the derived 27/88 — it is a form fitting sin$^2\theta$12 (a derived, divided quantity), not the primary matrix element.

Verdict — θ12: BANKED → IDENTIFIED (scorecard downgrade)

- The physically-primary form is the matrix element $|$U_e2$|^2$ = N_c/(rank·n_C) = 3/10 (0.06%) — 3 integers, matches the direct observable, mechanism-shaped. This is the form to carry.
- "sin$^2\theta$12 = 42/137 BANKED" is over-claimed: it fits sin$^2\theta$12 (0.14%) but sin$^2\theta$12 is $|$U_e2$|^2$/cos$^2\theta$13 (derived + divided), and 42/137 is a near-coincidence with the implied 27/88. It is not the matrix-element mechanism.
- Multiplicity is the red flag (Keeper's point): θ12 has ≥4 competing "canonical" forms (42/137, 4/13, 10/33, 27/88). A banked mechanism-forced value has ONE form. The multiplicity + the matrix-element resolution ▯ θ12 is IDENTIFIED (cleanest form $|$U_e2$|^2$=3/10, 0.06%), NOT BANKED.

Honest tier / discipline

- Solid: the physical observable is $|$U_e2$|^2$, its cleanest form is N_c/(rank·n_C)=3/10 (0.06%), 42/137 is a sin$^2\theta$12 near-fit. This DOWNGRADES the over-claim and identifies the matrix-element form.
- Held (Cal #27): I do NOT bank $|$U_e2$|^2$=3/10 either — it's the cleanest identified form (3 integers, matrix element, 0.06%), but the full PMNS mechanism (Engine B, the generation-state build) is what would forward-derive it. The cleanup's job is done: the "BANKED 42/137" over-claim is retired; θ12 is IDENTIFIED at the matrix-element form.
- @Keeper — for the scorecard v0.3: PMNS θ12 tier BANKED → IDENTIFIED, form $|$U_e2$|^2$ = N_c/(rank·n_C) = 3/10 (0.06%, matrix element); note 42/137 is a near-coincident sin$^2\theta$12 fit, not the primary form. No bank; graph current.

— Grace, 2026-07-14. $\theta 12$ cleanup: the physical observable $|U\_e2|^2$ = N_c/(rank·n_C) = 3/10 (0.06%, matrix element, mechanism-shaped) is the form to carry; $\sin^2\theta 12$ = 27/88 follows; "42/137 BANKED" is a near-coincident $\sin^2\theta 12$ fit and is over-claimed. $\theta 12 \rightarrow$ IDENTIFIED (scorecard downgrade), not BANKED. Multiplicity of competing forms confirmed. No bank.